

CSA1017 – SOFTWARE ENGINEERING

INDIVIDUAL REFLECTION REPORT

INTELLIGENT DISASTER EARLY WARNING AND EVACUATION MANAGEMENT SYSTEM

NAME: BATRAPALLI BINDHU

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ROLE: AI-Based Disaster Prediction and Risk Analysis

DESIGN / DEVELOPMENT DECISION:

I contributed to the AI-based prediction and risk-analysis component of the disaster-management system. My focus was on analysing disaster-related information such as rainfall, water levels, cyclone intensity, and affected regions to identify possible flood risks. I helped design a risk-classification approach that categorises areas into different severity levels and supports early warning decisions.

CHALLENGE FACED AND SOLUTION:

The main challenge was determining how multiple environmental factors could be combined to produce meaningful disaster-risk information. I addressed this by organising the inputs into important parameters and assigning suitable risk levels. I also considered the importance of providing early predictions rather than waiting until flooding becomes severe.

LEARNING GAINED:

I learned how artificial intelligence can support disaster prediction and decision-making. I understood that AI systems require reliable data and clearly defined parameters to generate useful results. I also learned how prediction systems can be connected with warning and evacuation modules.

SDG RELEVANCE:

The project supports SDG 11 by contributing to safer and more resilient communities. Early identification of flood risks can help authorities and citizens prepare before a disaster becomes critical.

COURSE OUTCOME CONTRIBUTION:

My contribution supports course outcomes related to software analysis, intelligent system design, requirements interpretation, and system integration. I learned how AI functionality can be incorporated into a practical software engineering solution.

POSSIBLE IMPROVEMENT:

I would improve the prediction module by using larger historical datasets, more advanced machine-learning models, continuous sensor data, and improved risk prediction techniques for different disaster conditions.